

The university of Azad Jammu and Kashmir ,
Muzaffarabad



Name	Nida Mughal
Roll no	2024-SE-41
Course Title	Computer networks
Lab	Switching and ARP
Submitted to	Dr. Asma javeid

Department of Software Engineering

Switching and ARP (MAC Address Learning)

Objective

The objectives of this lab are:

- To understand switching in computer networks.
- To learn how switches forward frames using MAC addresses.
- To understand the ARP (Address Resolution Protocol) process.
- To observe MAC address learning behavior in a switch.
- To learn how to check and clear the MAC address table.

Introduction

A switch is a Layer 2 networking device used to connect devices in a local area network (LAN). It forwards data frames using MAC addresses. When a switch receives a frame, it checks its MAC address table to determine the correct destination port.

ARP (Address Resolution Protocol) is used to map an IP address to a MAC address before communication takes place between devices. During communication, the switch learns MAC addresses automatically and stores them in the MAC address table.

In this lab, a simple network was created using two PCs and one switch to study switching and ARP behavior.

Devices Used

The following devices were used in this lab:

3 PCs

2 Switches

Copper Straight-Through Cables

Copper Cross-Over Cable

Network Topology

The network topology consists of:

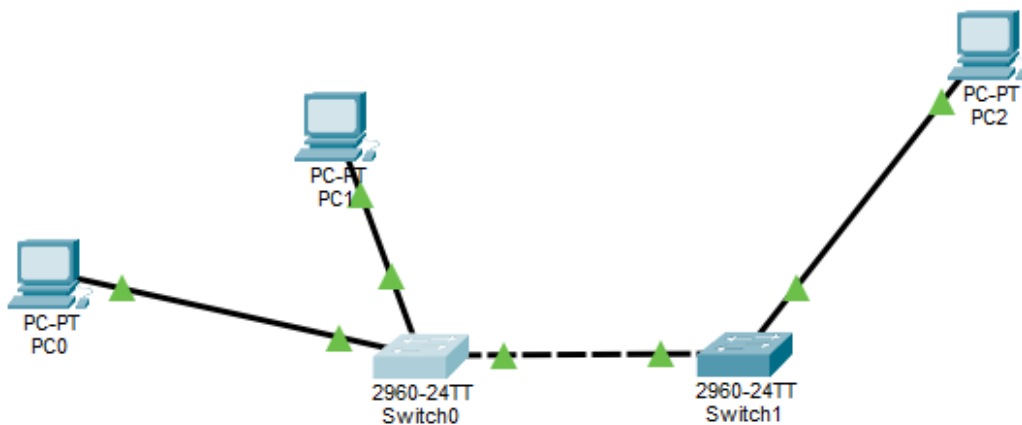
PC0 connected to Switch0

PC1 connected to Switch0

Switch0 connected to Switch1

PC2 connected to Switch1

PC-to-switch connections were made using straight-through cables, while switch-to-switch connection was established using a cross-over cable.



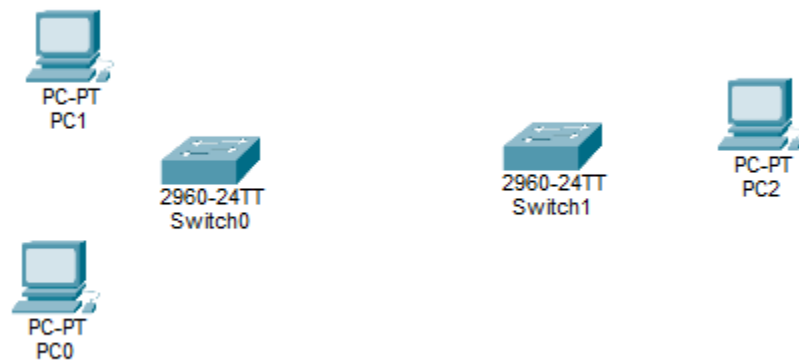
IP Addressing Table

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0

Procedure

Step 1: Create Network Topology

Cisco Packet Tracer was opened and a new workspace was created. Three PCs and two switches were added to the workspace.

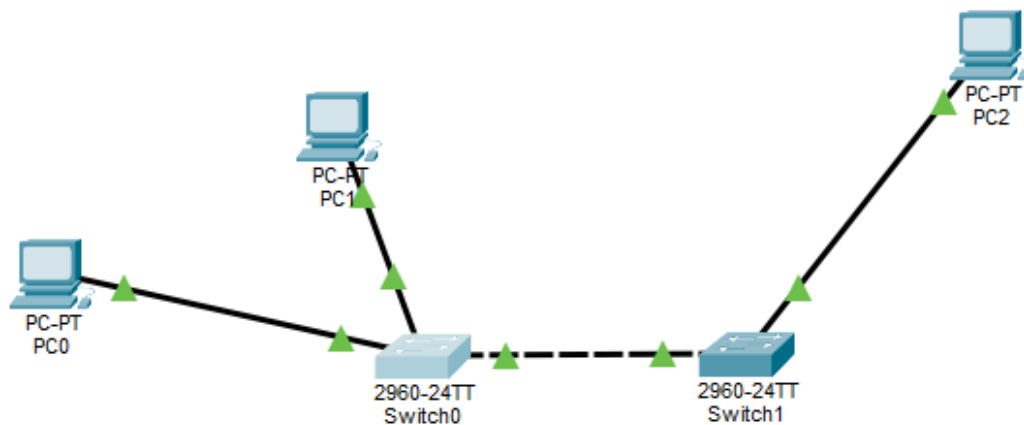


Step 2: Connect Devices

The devices were connected using Copper Straight-Through and Cross-Over cables.

Connections:

- PC0 to Switch0 using Straight-Through Cable
- PC1 to Switch0 using Straight-Through Cable
- Switch0 to Switch1 using Cross-Over Cable
- PC2 to Switch1 using Straight-Through Cable



Step 3: Configure IP Addresses

IP addresses were assigned to both PCs.

PC0 Configuration

- IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0

PC1 Configuration

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

PC2 Configuration

- IP Address: 192.168.1.3
- Subnet Mask: 255.255.255.0

IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

☐ DHCP	☒ Static
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

IPv4 Address	192.168.1.3
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

All systems were configured in the same network.

Step 4: Check MAC Address Table Before Communication

The switch CLI was opened and the following command was executed:

- show mac address-table

Initially, the MAC address table was empty because communication had not started yet.

```
Switch#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -

```

Step 5: Test Communication Using Ping

The Command Prompt of PC0 was opened and the following command was used:

- ping 192.168.1.2
- ping 192.168.1.3

The first ping request took some time because the ARP process was performed first. After ARP resolution, communication became successful.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=82ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 82ms, Average = 20ms

C:\>|
```

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Step 6: Verify MAC Address Table

The switch CLI was opened again and the following command was executed:

- show mac address-table

After successful communication, both switches learned the MAC addresses of connected devices automatically and stored them in their MAC address tables.

```

Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.978b.c3e0   DYNAMIC   Fa0/3
1       0001.c9e7.7223   DYNAMIC   Fa0/1
1       000a.4162.2b03   DYNAMIC   Fa0/3
1       000c.855d.beea   DYNAMIC   Fa0/2

Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.978b.c3e0   DYNAMIC   Fa0/1
1       0001.c9e7.7223   DYNAMIC   Fa0/3
1       000c.855d.beea   DYNAMIC   Fa0/3
1       0040.0b5e.e403   DYNAMIC   Fa0/3

```

Step 7: Clear MAC Address Table

The following command was used to clear dynamic MAC entries:

- clear mac address-table dynamic

After clearing the table, the show command was executed again to verify that the learned entries were removed.

```
Switch#clear mac address-table dynamic
```

Switch0:

```
Switch#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -

```

Switch1:

```
Switch#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0040.0b5e.e403    DYNAMIC Fa0/3

```

ARP Working Explanation:

ARP stands for Address Resolution Protocol. It is used to find the MAC address of a device when its IP address is known.

The ARP process works as follows:

1. The sender device checks whether the destination MAC address is available.
2. If the MAC address is unknown, the sender broadcasts an ARP request.
3. The destination device replies with its MAC address.
4. The sender stores the MAC address in its ARP table.

5. Communication starts successfully.

MAC Address Table Explanation

The switch stores learned MAC addresses in a MAC address table, also known as the CAM table. Whenever a device sends frames through the switch, the switch learns the source MAC address and associates it with the connected port.

This process helps the switch forward frames efficiently without flooding unnecessary traffic.

Commands Used

- Show MAC Address Table
- show mac address-table

Purpose: Used to display learned MAC addresses on the switch.

- Clear Dynamic MAC Table
- clear mac address-table dynamic

Purpose: Used to remove dynamically learned MAC addresses from the switch.

- Ping Command

ping 192.168.1.2

Purpose: Used to test communication between two devices.

Observations

- Initially, the MAC address tables were empty.
- After pinging, both switches learned MAC addresses automatically.
- ARP resolved IP addresses into MAC addresses.
- Communication between all PCs became successful.
- Switches forwarded frames between networks correctly.
- The MAC address tables were cleared successfully using the clear command.

Result:

The lab was completed successfully. Switching behavior, ARP process, and MAC address learning were observed and verified using switch commands.

Conclusion:

In this lab, an extended LAN network was created using three PCs and two switches. IP addresses were configured successfully, and communication between all systems was tested using ping commands. The switches learned MAC addresses dynamically and stored them in their MAC address tables. The ARP process, MAC learning behavior, and frame forwarding between switches were successfully observed. The MAC address tables were also cleared and verified successfully.